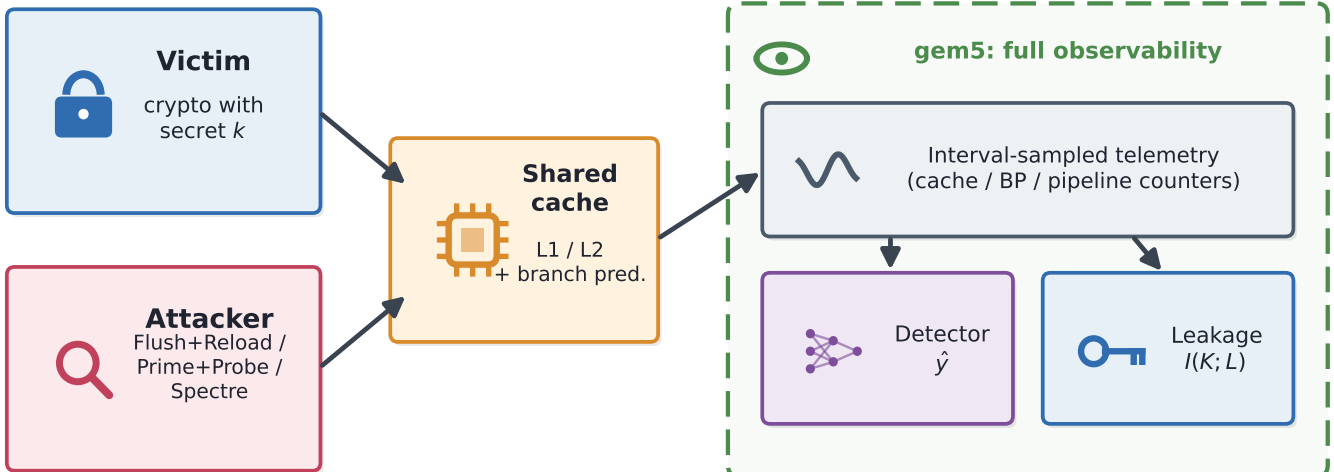
Victim
crypto with secret k
Attacker
Flush+Reload / Prime+Probe / Spectre
Shared cache
L1 / L2 + branch pred.
gem5: full observability
Interval-sampled telemetry (cache / BP / pipeline counters)
Detector $\hat{y}$
Leakage $I(K;L)$